Why Does CBT Fail? A Critical Review of Ten Proposed Explanations, with Executive Function and Adult ADHD as a Test Case

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Abstract

Non-response to cognitive behavioral therapy (CBT) does not identify a single failure of treatment, theory or delivery. Evidence that a treatment package works does not establish which components cause improvement or why some patients do not benefit. We critically evaluate ten proposed reasons CBT may fail, using selected meta-analyses, component trials and mechanistic studies rather than claiming systematic coverage. The review covers efficacy, mechanisms, patient and context factors, and delivery. Executive function and adult attention-deficit/hyperactivity disorder provide the deepest case study. CBT remains effective on average, but benefits depend on the comparator, and many limitations are shared with other treatments. Component findings are mixed and often underpowered; they establish neither the universal necessity nor the irrelevance of cognitive restructuring. For example, a depression review (16 studies, 22 comparisons) favoured full treatments over versions with one component removed ($g = 0.21$, 95% confidence interval $[0.03, 0.39]$), despite substantial methodological limitations. Evolutionary conservation motivates learning-based hypotheses but cannot rank therapies or establish that experience must precede cognition. Researcher allegiance and publication selection warrant scrutiny without establishing CBT-specific reviewer suppression. Acceptance, alliance, reactance and feedback research further qualify accounts that isolate technique from goals and delivery. We propose falsifiable component and moderator tests of which interventions produce durable functional change, for whom and under what conditions.

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1. Introduction

Cognitive behavioral therapy (CBT) is among the most heavily tested psychotherapies. A meta-analysis of 41 placebo-controlled trials in anxiety and related disorders (total $N = 2{,}843$) found that CBT outperformed placebo on target symptoms with a moderate effect (Hedges' $g = 0.56$) (Carpenter et al., 2018). The same literature also shows a ceiling: across 87 studies only about half of patients respond to CBT for anxiety disorders, and the figure depends on how response is defined (Loerinc et al., 2015); a larger meta-analysis of 100 articles reports a mean remission rate of 51.0% (Springer et al., 2018). A therapy can work on average and still leave open *why* it works and *for whom*.

The textbook account of CBT's cognitive component is a top-down loop. A situation triggers an automatic thought ("I am going to fail this talk"); the patient records it, names the distortion, weighs evidence for and against, and generates a more balanced thought; the feeling is expected to follow. This review first maps ten reasons given for why CBT fails (Section 3) and then grades the evidence for each. Its deepest case study is a narrow question about that loop: *does the mechanism ask the patient to use a capacity that anxiety, and in some patients attention-deficit/hyperactivity disorder (ADHD), has already degraded?* This motivates tests of whether executive capacity and stress moderate cognitive skill use, and whether components with different executive demands produce different outcomes. Executive demand alone does not establish those clinical differences.

The argument has a popular version. Judson Brewer, a clinician-researcher, argues that anxiety takes the prefrontal cortex "off-line," so that willpower and reasoning are unavailable at the moments when they are needed (Brewer, 2016, 2021), and he has applied this directly to CBT (Brewer, 2026). Our aim is to find out how much of this survives contact with the primary literature. Our answer, in brief, is that a narrow version survives and the strong version does not. The same standard is applied to the other reasons: dropout and non-response (Section 12), patient and context factors (Section 13), therapist and delivery factors (Section 14), and the theory, measurement and evidence-quality critiques (Section 15).

What we find. (1) Reappraisal does draw on working memory in laboratory studies, but evidence that acute stress abolishes it is thin and contested. (2) The direct evidence that executive function moderates CBT outcome is sparse and mostly null for overall outcome, with the clearest signal at the level of the cognitive skill itself. (3) Component findings are mixed: several trials detect no added benefit, whereas a depression component review favours full packages and some direct comparisons favour cognitive therapy; many designs are underpowered. (4) CBT's headline effects shrink with active controls and bias adjustment; whether they shrink over time is disputed. (5) In adult ADHD, CBT is modestly effective, mostly as an adjunct to medication, and the plausible active ingredient is behavioural compensation; we found no trial testing ADHD as a moderator of CBT for anxiety in adults. (6) Beyond that argument, dropout (about 16% before and 26% during CBT), non-response, relapse and under-recorded harm are real and quantified, but mostly shared with other psychological treatments; patient, context and delivery factors shift outcomes without isolating a CBT-specific failure; and mechanism, measurement and trial-quality limitations constrain causal interpretation and effect estimates.

Contributions. (i) An explicit decomposition of cognitive restructuring into executive demands (Section 4); (ii) an evidence-graded review of the stress-and-prefrontal premise, including a check of the "off-line" claim against its sources (Section 5); (iii) a review of efficacy, evidence selection and component evidence (Sections 7–8); (iv) a review of alternatives and of CBT in adult ADHD (Sections 9–10); (v) the direct moderator evidence (Section 11); (vi) a graded map of ten reasons CBT may fail (Section 3) and reviews of non-response, patient, delivery and mechanism factors (Sections 12–15); (vii) an analysis of evolutionary learning and clinical goals and engagement, with explicit limits on the inferences they support (Sections 15.1 and 13.1); (viii) testable hypotheses (Section 17).

2. Method

This is a critical theoretical review, not a systematic review reported in full under the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. We select literatures relevant to ten reasons CBT may fail, with executive function and adult ADHD as the deepest case study. The initial searches ran in October 2026 across PubMed, Europe PMC,

Crossref, publisher pages and reference lists of key meta-analyses. They covered efficacy and bias; components and mechanisms; executive function and stress; ADHD; non-response, dropout, relapse and harm; patient and context factors; delivery; and theory and measurement. We prioritised meta-analyses, component trials, moderator studies, and critiques and defences of CBT. A targeted extension in November 2026 examined evolutionary learning, researcher allegiance and publication selection, acceptance, engagement, alliance and intervention timing. This extension followed proposed claims and relevant counterevidence; it was not an exhaustive update of every literature.

Verification. For the extension, we checked the title, authors and publication year of scientific references against DOI-resolved Crossref records. We distinguished this bibliographic check from reading abstracts or relevant full-text passages to verify findings. Quoted numbers came from those passages, not from search summaries or memory. The public source notes record reading levels, scope limits, inaccessible candidates and retrieval provenance; copyrighted reading copies remain outside the public repository. Earlier source notes and the public verification log document prior DOI checks and abstract reading; their partial-verification flags remain in place. Popular sources motivate questions but do not establish empirical findings.

Limitations of the method. Selection was not independently duplicated, the review was not preregistered, and we did not formally assess the risk of bias of every included review. Our synthesis therefore remains vulnerable to selection and interpretation bias, including our own preference for the proposed argument. Evidence from depression is used where anxiety evidence is thin, with the distinction stated. A structured umbrella search with independent screening and risk-of-bias assessment remains necessary before claiming systematic coverage (Section 17).

3. A map of the reasons CBT fails

"CBT does not work" is not one claim. It bundles at least ten different ones, which differ in what they say fails (the average effect, the individual response, the mechanism, the evidence or the delivery), in how well they are supported, and in whether they are specific to CBT at all. Table 1 lists them with the verdict this review reaches and the section that holds the evidence. We use four verdicts: **supported** (consistent evidence, quantified), **partly supported** (real but smaller or more conditional than the popular version),

contested (credible evidence on both sides) and **open** (little or no direct evidence).

Two cautions apply throughout. First, many of the best-documented limits (dropout, relapse, comorbidity, harm under-reporting, therapist variability) are limits of psychological treatment in general, not of CBT in particular; we say so each time the source does. Second, a limit on certainty is not evidence of ineffectiveness: CBT keeps a moderate average effect under almost every sensitivity analysis we found (Plessen et al., 2023; Carpenter et al., 2018). The map is therefore a list of *where and why* CBT falls short of its reputation, not a verdict that it fails.

The executive-function argument (R1) is the review's deepest case study, because it makes a testable, mechanism-level prediction and because it was the starting point of the project. It is one reason among ten, and Sections 12–15 treat the others at the same standard of evidence.

4. What cognitive restructuring asks of the patient

Consider the standard thought-record exercise. The patient must (a) *notice* an automatic thought while it is occurring, (b) hold the thought, the supporting evidence, the contradicting evidence and a candidate alternative in mind at once, (c) suppress the pull of the automatic thought while evaluating it, and (d) shift from being inside the feeling to observing it. Mapped onto the best-known model of executive function, these are three separable components: updating and holding information (working memory), inhibiting a prepotent response, and shifting between mental sets (Miyake et al., 2000).

Laboratory evidence that reappraisal draws on executive resources. In event-related-potential studies, high working-memory load eliminated the effect of reappraisal on the late positive potential (Gan et al., 2017; Adamczyk et al., 2022). Working-memory capacity predicted reappraisal success (Schmeichel et al., 2008), and reappraising late in an emotional response, once emotion is already intense, costs self-control resources (Sheppes and Meiran, 2008). People also choose differently as intensity rises: they prefer distraction to reappraisal in high-intensity negative situations and reappraisal in low-intensity ones (Sheppes et al., 2011).

Evidence that it works, and can be trained, when load is low. Instructed reappraisal reduces negative experience in the laboratory (Hofmann et al., 2009) and can be trained (Denny and Ochsner, 2014). So the claim is not

	Reason	Verdict	Where
R1	The cognitive skill demands executive function that anxiety (and ADHD) degrade	contested; direct moderator evidence open	Sections 4–6, 11
R2	The incremental benefit of cognitive restructuring is inconsistent	contested; depends on disorder and comparison	Section 8
R3	Headline effects are inflated by waitlists, bias, low-quality trials and measurement choices	supported	Sections 7, 15
R4	Many patients do not respond or drop out	supported (not CBT-specific)	Section 12
R5	Gains erode after treatment ends	partly supported	Section 12
R6	Harms and deterioration are under-recorded	supported for reporting; CBT lowers deterioration versus controls	Section 12
R7	Patient factors (personality disorder, chronicity, adversity, medical comorbidity, neurodivergence)	partly supported; mostly not CBT-specific; autism evidence thin	Section 13
R8	Context: WEIRD trials, culture, waiting times, deprivation, access	supported	Section 13
R9	Delivery: therapist variability, drift, weak training, dose, format	partly supported; some causal evidence	Section 14
R10	The theory of change is under-tested; common factors may explain more	contested	Section 15

Table 1: **CBT's limitations concern different outcomes and levels of explanation.** Ten proposed reasons, with the verdict of this critical review.

that reappraisal fails; it is that it *costs* executive resources, which matters when those resources are scarce.

Two clarifications. First, this is a task analysis plus laboratory support, not a treatment finding. Clinicians scaffold the load (worksheets, a therapist who prompts for evidence, graded practice). Second, executive function is not a single resource; the argument concerns working memory, inhibition and shifting, not general intelligence. We return to both in Section 16.

5. Stress, anxiety and the prefrontal executive system

5.1. Mechanistic neuroscience

The most developed account is due to Arnsten. Her review of stress signalling describes the prefrontal cortex (PFC) as the most evolved brain region and also the one most sensitive to the detrimental effects of stress exposure (Arnsten, 2009). The proposed mechanism is an inverted-U in catecholamine signalling: moderate norepinephrine and dopamine levels support prefrontal network firing and working memory, while the high levels of uncontrollable stress weaken them and strengthen amygdala and striatal habit circuitry. A later review summarises the result as stress rapidly flipping the brain from reflective to reflexive control of behaviour (Arnsten, 2015). Acute uncontrollable stress impairs PFC-dependent working memory in monkeys and rodents (Arnsten and Goldman-Rakic, 1998). In humans, a meta-analysis finds that acute stress impairs working memory and cognitive flexibility, with more nuanced effects on inhibition and many moderators (Shields et al., 2016), and imaging suggests a reallocation of resources toward salience networks rather than a switch-off (Hermans et al., 2014).

Where the "off-line" language comes from. The phrase traces to a hedged line in a 1998 monkey study ("Stress may take the PFC 'off-line"'; Arnsten and Goldman-Rakic 1998), and its author still uses it in quotation marks for *uncontrollable* stress (Arnsten, 2020). Brewer and Roy apply it to worry and cite Arnsten's 2009 and 2015 reviews, which are not studies of worry (Brewer and Roy, 2021); their app-based trial ($n = 65$, treatment-as-usual control) did not test a prefrontal mechanism (Roy et al., 2021). We therefore read "offline" as a popularisation of *impaired, in a graded and dose-dependent way*, mainly under uncontrollable stress.

5.2. Does stress abolish reappraisal in humans?

Evidence for this specific link is contested. The founding study found that participants stressed before learning showed no fear reduction after regulation training (Raio et al., 2013), and a review summarises that acute stress may impair the persistent inhibition of fear, presumably by altering prefrontal function (Raio and Phelps, 2015). Five later studies ($n = 54$–98) found no group-level impairment (Shermohammed et al., 2017; Sandner et al., 2021; Langer et al., 2021, 2023; Hamza et al., 2024), though two reported sex- or cortisol-linked effects. Their laboratory stressors were moderate, so the null results may not transfer to clinical-level arousal; that gap is itself an open question.

5.3. Cognitive psychology of anxiety

Attentional Control Theory proposes that anxiety impairs the inhibition and shifting functions of executive control, so that anxious individuals are more captured by threat and worse at redirecting attention (Eysenck et al., 2007). The theory predicts that *efficiency* (effort and time) suffers more than *effectiveness* (accuracy), because anxious people compensate with extra effort. Meta-analytic support exists at group level ($g = -0.58$; Shi et al., 2019) and for working memory ($g = -0.334$; Moran, 2016), but with qualifiers: a meta-analysis of 32 studies ($N = 13{,}084$) found generalised anxiety disorder (GAD) associated with poorer cognitive flexibility and working memory but not inhibitory control, and with no association with any executive domain in older adults (Nguyen et al., 2025). Working memory was unrelated to social anxiety in one study (Harrison et al., 2025), and one study found *enhanced* control over negative distractors in GAD (Hallion et al., 2019).

6. Executive function in ADHD

ADHD is heterogeneous: executive difficulties are common but not universal. A meta-analysis of 83 studies found that groups with ADHD underperform controls on every executive task class examined, most strongly on response inhibition, vigilance, working memory and planning, with medium effect sizes (0.46–0.69) (Willcutt et al., 2005). The same authors caution that executive weaknesses are neither necessary nor sufficient to cause all cases of ADHD, and fewer than half of the (mostly child) samples show impairment on any single executive test (Willcutt et al., 2005). Measured more carefully, impairment is more prevalent: in a sample of 136 children, 89% of those with ADHD

were impaired on at least one executive function (62% working memory, 27% inhibitory control, 38% set shifting) (Kofler et al., 2019).

Anxiety is common in adult ADHD: a meta-analysis of 58 studies (N = 18,821) found higher anxiety scores in adults with ADHD than without ($g = 0.77$; $g = 0.95$ against non-clinical and $g = 0.44$ against clinical controls) (Hoskins et al., 2026). Whether anxiety adds executive load in ADHD is less clear: in a sample of 339 children, trait anxiety did not uniquely predict working memory or inhibitory control, nor moderate the ADHD effect (Marsh et al., 2024). The stacking assumption (low baseline executive capacity, lowered further by anxiety) is therefore plausible but, in children at least, not supported by the one careful test we found.

7. What the efficacy evidence shows

7.1. The case for CBT

Against placebo, CBT outperforms on anxiety target symptoms ($g = 0.56$ across 41 trials; Carpenter et al. 2018). Against inactive controls, effects are larger: a meta-analysis of 144 trials found $g = 0.75$ (depression) to 0.88 (social anxiety disorder), with 0.80 for GAD and 0.81 for panic disorder (Cuijpers et al., 2016). A 2025 network of 375 trials gave anxiety-spectrum effects between 0.5 and 1.0, above 1.0 for PTSD and specific phobia (Cuijpers et al., 2025). In social anxiety, individual CBT ranked best of the therapies compared in a network meta-analysis of 101 trials (Mayo-Wilson et al., 2014), and in GAD only CBT remained significantly better than treatment as usual at 3–12 months in a network of 65 trials (Papola et al., 2024). Any critique has to explain why CBT works at all.

7.2. The headline shrinks with better controls and bias adjustment

- **Control type.** More than 80% of anxiety trials used waitlist controls, and the few that did not showed much smaller effects (Cuijpers et al., 2016); only 17.4% of trials in that analysis were high quality. In the 375-trial analysis, waitlist-controlled effects exceeded $g = 0.94$, while care-as-usual comparisons gave 0.22–1.13 (Cuijpers et al., 2025).

- **Bias.** In depression, psychotherapy effects fell from $g = 0.70$ to $g = 0.31$ after adjusting for several sources of bias, with only 23% of studies at low risk of bias (Cuijpers et al., 2019b). In a cohort of National Institutes of Health-funded depression trials, 13 of 55 initiated grants did not lead to

publication (Driessen et al., 2015). Adding six unpublished studies to 20 published control comparisons changed the pooled effect from $g = 0.52$ (95% CI $[0.37, 0.68]$) to 0.39 ($[0.08, 0.70]$). The reported 25% decrease is a change in point estimates; the published–unpublished contrast was uncertain ($p = .07$), and the pooled treatment effect remained positive. An umbrella review of 102 recent meta-analyses (3,782 RCTs) found most target-symptom effects small and pooled psychotherapy effects of 0.34 against placebo or treatment as usual (Leichsenring et al., 2022).

- **Response and remission.** About half of anxiety patients respond or remit (Loerinc et al., 2015; Springer et al., 2018).

7.3. Evidence volume, allegiance and reviewer selection

Evidence volume, comparative efficacy and component causality are different questions. A large literature can increase confidence in a treatment, but its size alone establishes neither superiority to active alternatives nor which component causes improvement. Researcher allegiance—investigators' preference for a treatment—was associated with comparative outcomes across 30 meta-analyses (Munder et al., 2013). That association warrants scrutiny, but does not alone separate bias from differences in treatment efficacy, expertise or implementation. Nor does allegiance adjustment necessarily eliminate a CBT advantage: one review retained an advantage after adjustment, with superiority over psychodynamic but not interpersonal or supportive therapies (Tolin, 2010). These findings justify examining particular comparisons, not declaring all therapies equivalent.

Publication bias must also be distinguished from reviewer preference. Among the 13 unpublished studies identified by Driessen et al. (2015), two were known to have been submitted and rejected, six were never submitted, and submission status was unclear for five. This is not evidence that CBT-oriented reviewers excluded competing therapies. A randomized experiment did find that peer reviewers favoured a positive-result manuscript over an otherwise comparable no-difference version, but it involved orthopaedic journals, not psychotherapy (Emerson et al., 2010). We did not identify direct evidence establishing CBT-specific reviewer suppression in the sources examined for this extension. Testing that hypothesis requires reviewer-orientation data and comparable manuscripts, with non-submission, editorial decisions and review recommendations distinguished. Prospective registration, complete reporting, credible active comparators and independent replication address

evidence selection without presuming that one therapeutic orientation is uniquely biased.

7.4. Does the effect shrink over time?

This is contested, and the evidence is mixed. In depression, one meta-analysis reported a steady decline in CBT effects since its introduction (Johnsen and Friborg, 2015), but a reanalysis restricted to randomised comparisons judged the decline "most likely a spurious finding" (Cristea et al., 2017), and a 409-trial meta-analysis found no reduction by publication year (Cuijpers et al., 2023). In anxiety, a meta-regression of 49 controlled studies over 30 years found a flat trend (publication year $B = -0.008$, $p = .24$): effects "did not increase over the last 3 decades" (Hofmann et al., 2025). Placebo-controlled anxiety estimates have been reported at lower values in successive reviews (0.73, 0.56, 0.24), but those come from separate reviews with different case mixes, not from one meta-regression. We therefore do *not* claim a decline over time; the defensible statement is that effects have not grown and that they are sensitive to control condition and bias.

8. Where the active ingredients may lie

8.1. Component and dismantling evidence

The foundational depression dismantling trial ($N = 150$) detected no advantage of complete cognitive treatment over behavioural activation, with or without automatic-thought skills, at termination or six months (Jacobson et al., 1996). A larger depression trial ($N = 440$) established behavioural activation's non-inferiority to CBT on the Patient Health Questionnaire-9 (PHQ-9) at 12 months (Richards et al., 2016). The modified intention-to-treat difference was 0.1 PHQ-9 points (95% CI $[-1.3, 1.5]$), within the trial's prespecified 1.9-point margin; the per-protocol analysis supported the same inference. Behavioural activation was delivered by junior mental health workers and CBT by psychological therapists, so this pragmatic trial compared treatment packages and workforces rather than isolating one component. It supports a viable treatment without an explicit restructuring module, not identical effects or a particular learning mechanism. In anxiety:

- In a 108-trial meta-analysis, cognitive therapy, exposure and their combinations were efficacious with "no differential efficacy for any treatment components" (Norton and Price, 2007).

- In a full-factorial trial of internet CBT for social anxiety ($N = 464$), adding psychoeducation or exposure improved post-treatment outcomes; adding cognitive restructuring or attention training did not show superiority (Šipka et al., 2025). The psychoeducation and exposure advantages were no longer significant four months after treatment, as conditions including cognitive restructuring or attention training improved further. Substantial attrition limits the follow-up interpretation. The exposure component itself included recording and comparing expected and observed outcomes, so it did not exclude cognitive processing.

- Adding cognitive restructuring to prolonged exposure did not enhance outcome in assault survivors with PTSD (Foa et al., 2005), nor in children with social anxiety (de Mooij et al., 2023).

- A component network meta-analysis in panic disorder (72 studies) singled out interoceptive exposure, not cognitive restructuring, as efficacy-enhancing (Pompoli et al., 2018).

- In GAD, the network meta-analysis authors concluded that pure cognitive restructuring was not a decisively active ingredient when considered alone (Papola et al., 2024); the paper was later corrected for a duplicated cohort and an error in standard-error handling (JAMA Psychiatry, 2024).

8.2. Evidence against the claim

These findings are not uniform.

- Cognitive therapy beat exposure in social phobia in a meta-analysis of 20 direct comparisons ($N = 1{,}308$), with no difference in PTSD, OCD or panic (Ougrin, 2011).

- In a trauma trial, adding cognitive restructuring to exposure was associated with PTSD rates of 31% versus 63–75% without it at 6 months (Bryant et al., 2008).

- In-session cognitive restructuring correlates with outcome ($r = .35$), though from only four studies and correlationally (Ezawa and Hollon, 2023).

- In a single-session trial with elevated public-speaking anxiety ($N = 249$), restructuring before exposure and behavioural experiments both outperformed exposure without explicit expectancy processing (Johnco et al., 2025). The small, non-significant difference in anxiety reduction favoured behavioural experiments, which also produced greater expectancy change in a shorter session. Follow-up was limited to one week.

- The exposure–cognition split is blurred: exposure may itself work by changing the expectancy of harm (Hofmann, 2008).

8.3. Why the component evidence is weak on both sides

A review of 16 depression component studies found a small pooled advantage for full treatments over versions with one component removed ($g = 0.21$, 95% CI $[0.03, 0.39]$) (Cuijpers et al., 2019a). The components differed, so this is not an estimate of cognitive restructuring's unique contribution. Only one study had low risk of bias, and all but one lacked power to detect effects smaller than $g = 0.55$. Mediation evidence is correlational: cognitive change tracks symptom change however it is produced (Cristea et al., 2015; Lorenzo-Luaces et al., 2015), cognitive mediation of symptom reduction was limited in a depression cohort (median $r = .06$; Vittengl et al., 2014), and reviewers say no common or specific factor meets causal criteria (Kazdin, 2007; Cuijpers et al., 2019c). A null component result is thus weak evidence that restructuring adds nothing, and a positive correlation is weak evidence that it matters.

8.4. The strongest defence: practice when calm

A defender can answer that patients rehearse when calm rather than undertake effortful restructuring during peak distress. We accept the force of this possibility, but successful rehearsal would not by itself identify practice or habit formation as the active ingredient. It could improve retrieval, cognitive skill, expectancies or behavioural routines; these alternatives require measurement and component tests. Repeated performance does not reliably produce outcome-insensitive habits in human laboratory tasks (de Wit et al., 2018). Exposure protocols emphasise learning through expectancy violation (Craske et al., 2014). The relevant questions are what is rehearsed, what changes, and whether the acquired response transfers to situations with high executive demands.

9. Alternatives and their evidence

9.1. Mindfulness and acceptance-based treatments

Mindfulness and acceptance-based approaches train attention and responses to thoughts and feelings rather than relying exclusively on disputing thought content. Acceptance-based work overlaps with the broader CBT family (Hayes et al., 2006). Its inclusion here does not establish an absence of cognitive or executive demands. Brewer frames anxiety as a habit loop (trigger, worry, a feeling of control) and proposes noticing, curiosity about the reward value of worry, and a "bigger, better offer" (Brewer, 2021). Evidence for the specific programme is early: in a randomised trial of an app-based programme for GAD ($n = 65$), anxiety scores fell 67% versus 14% under usual care, with mediation implicating reduced worry (Roy et al., 2021). The trial is small, self-reported, compared with usual care rather than CBT, and has the programme's developer among its authors.

The following syntheses do not establish that mindfulness and acceptance approaches outperform evidence-based comparators. A meta-analysis found essentially no difference ($g \approx 0.00$) between mindfulness-based interventions and evidence-based treatments, often CBT (Goldberg et al., 2018); another found no difference from traditional CBT or behavioural therapies ($g = -.07$, nine comparisons; Khoury et al. 2013). For acceptance and commitment therapy (ACT) versus CBT, the pooled difference was non-significant in one meta-analysis (A-Tjak et al., 2014), and a sceptical reading of 60 RCTs concluded that ACT is not yet well-established for any disorder (Öst, 2014). In 276 adults with anxiety disorders, eight weeks of mindfulness-based stress reduction was noninferior to escitalopram, with fewer discontinuations for adverse events (Hoge et al., 2023). That is a non-inferiority result against a first-line drug, not evidence of superiority to CBT. These comparisons do not establish universal equivalence across disorders, outcomes and follow-up periods. Similar outcomes can arise through shared or different processes; they do not identify a mechanism by themselves.

Reward revaluation as a hypothesis. An alternative self-regulation account proposes that attending to the experienced consequences of an action can change its reward value and reduce reliance on effortful restraint (Ludwig et al., 2020). This is a mechanistic proposal, not comparative evidence against CBT. An app-based eating study linked model-estimated reward values to reported eating, but had no control group (Taylor et al., 2021). Its larger

naturalistic sample showed decreasing estimated reward values without a corresponding decrease in reported eating behaviour across tool uses. The study supports further testing of reward revaluation, not a causal conclusion about the intervention, a general mechanism for anxiety, or superiority to cognitive restructuring.

9.2. Medication in ADHD

In a network meta-analysis of 133 double-blind trials, amphetamines (standardised effect -0.79) and methylphenidate (-0.49) outperformed placebo in adults on clinician ratings, over short follow-up (Cortese et al., 2018). The certainty varied from high or moderate to low or very low. A Cochrane review of extended-release methylphenidate found a self-rated effect of -0.37, judged 20 of 21 trials at high risk of bias, and noted that all trials excluded participants with psychiatric comorbidity such as anxiety or selected previous responders (Boesen et al., 2022). So the pivotal medication trials say little about adults with ADHD *and* anxiety.

10. CBT in adult ADHD

10.1. Efficacy

A randomised trial of 86 medication-treated adults found that CBT outperformed relaxation with educational support, with 53% versus 23% responders on clinician global ratings (Safren et al., 2010); a smaller earlier trial found 56% versus 13% responders for CBT added to medication (Safren et al., 2005). A meta-analysis of 32 studies found a between-group effect of $g = .65$ on symptoms, smaller with active controls, and neither medication status nor treatment length moderated outcome (Knouse et al., 2017). Another found SMD 0.76 against waitlist but 0.43 against active controls (Young et al., 2020), and a 2026 meta-analysis of 14 RCTs found SMD -0.45 on core symptoms and -0.43 on executive function (Liu et al., 2026). A component network meta-analysis of 113 RCTs found CBT better than placebo only on clinician-reported measures (Ostinelli et al., 2025), and blinded-rater effects of psychological treatments were attenuated to non-significance in an earlier synthesis that did not separate adult from child samples (Sonuga-Barke et al., 2013). An umbrella review judged CBT, atomoxetine and methylphenidate to have at least moderate-certainty evidence in adults, with no high-certainty long-term evidence for any intervention (Gosling et al., 2025). The Cochrane review rates the CBT evidence as low quality (Lopez et al., 2018).

10.2. What the protocols contain, and where CBT sits

The effective protocols are skills-heavy: time management, organisation and planning, with adaptive thinking as one module (Safren et al., 2010; Solanto et al., 2010). One review argued that learning behavioural compensatory strategies "may be a critical active ingredient" (Knouse and Safren, 2010), and the authors of one trial proposed behavioural control as the key mechanism (Young et al., 2017). We found no dismantling study of cognitive restructuring in adult ADHD, so this is inference. In the largest multi-centre trial ($N = 433$), group CBT did not beat structured clinical management, while methylphenidate beat placebo (Philipsen et al., 2015). Guidelines position CBT as an adjunct to medication (Kooij et al., 2019; National Institute for Health and Care Excellence, 2018). Results on adding medication to CBT are mixed: no incremental benefit in a small double-blind pilot (Weiss et al., 2012), but greater early improvement in an unblinded trial ($N = 88$), with the gap narrowing over follow-up (Cherkasova et al., 2020).

10.3. Anxiety in ADHD

In a meta-analysis of 13 RCTs ($n = 744$), CBT-based interventions reduced anxiety in ADHD modestly (SMD -0.29 post-treatment, -0.42 at follow-up), with effects more apparent in adults; most trials delivered ADHD-focused rather than anxiety-targeted CBT (Wang et al., 2026). Another meta-analysis found CBT improved anxiety, particularly at follow-up, predicted by core-symptom reduction (López-Pinar et al., 2020). In children with primary anxiety in group CBT ($N = 842$), neither ADHD diagnosis nor subtype predicted response or remission, which cuts against the strong form of our thesis (Gould et al., 2018). The adult version of that question appears untested.

11. Does executive function moderate CBT outcome?

This is the question on which the argument stands or falls. The direct evidence is thin and mostly from older adults.

- **Supports (at the level of the skill).** In older adults, poorer cognitive flexibility predicted poorer cognitive-restructuring quality and less distress reduction from restructuring, but *not* overall CBT outcome: those with poorer flexibility "can still benefit from standardised CBT" (Johnco et al., 2014).

- **Supports (pilot).** In a pilot of CBT for GAD in older adults, participants whose executive scores remained low did not respond, while those whose scores improved responded as well as an intact-executive group (Mohlman and Gorman, 2005).

- **Null or mixed.** No baseline executive effect was found in adult OCD (Vandborg et al., 2016), child anxiety (Godovich et al., 2020) or brief psychotherapy for late-life depression (Schurr et al., 2025); a systematic review of late-life anxiety found no replicated executive predictor (Kyriacou et al., 2023). In a meta-analysis of antidepressant response in late-life depression, only planning and organisation predicted outcome (Pimontel et al., 2016).

- **Opposite direction.** Better executive performance went with non-response to exposure-based CBT in paediatric OCD (Hybel et al., 2017), and worse set-shifting went with better CBT response in late-life depression (Goodkind et al., 2016).

- **Adaptation helps.** Problem-solving therapy beat supportive therapy in older adults with executive impairment, with a larger advantage at greater impairment (Areán et al., 2010), which points to adapting therapy rather than abandoning it.

Taken together, "low executive function means CBT fails" is not supported. A narrower claim is: executive capacity may limit uptake of the *cognitive skill* without limiting overall response, because CBT also works through routes that do not need it. We found no adequately powered study of working-age adults with ADHD and anxiety.

12. Non-response, dropout, relapse and harm

Dropout. Dropout is the most visible failure. A meta-analysis of 115 CBT studies (20,995 participants) found 15.9% dropout before treatment and 26.2% during it, highest in depression and in e-therapy (Fernandez et al., 2015). CBT is not unusual here: across 669 psychotherapy studies the weighted dropout was 19.7% and was not moderated by therapy orientation (Swift and Greenberg, 2012); in 54 depression trials it was 17.5% in psychotherapy arms (Cooper and Conklin, 2015), and in routine-care CBT for depression about 25% (Hans and Hiller, 2013). CBT dropout is, if anything, lower

than medication dropout (Cuijpers et al., 2013a). Reported predictors are a weak alliance ($d = .55$; Sharf et al., 2010), comorbid personality disorder and minority status (Cooper and Conklin, 2015), and, in self-guided online programmes, male sex, low education and comorbid anxiety (Karyotaki et al., 2015).

Non-response. Even under trial conditions many patients do not respond. In a moderate-to-severe depression trial, 16-week response was 58% and remission 40% for cognitive therapy, with outcomes depending on therapist experience at one site (DeRubeis et al., 2005). Across 92 psychotherapy studies fewer than half of patients met response or remission criteria, and 48% of care-as-usual patients also no longer met depression criteria, so the added value was about 14 percentage points (Cuijpers et al., 2014). For anxiety, one review puts non-response at about a third of completers and finds that "no predictor has been consistently supported" (Taylor et al., 2012); see also Loerinc et al. (2015) and Springer et al. (2018). Routine services do less well than trials: of 19,395 English IAPT patients, 40.3% reliably recovered, 63.7% reliably improved and 6.6% reliably deteriorated (Gyani et al., 2013). The prognostic factors with support (comorbid personality disorder, chronicity, adversity) apply to all treatments (Section 13); baseline severity did not moderate CBT versus medication in an individual-participant meta-analysis of 1,700 patients (Weitz et al., 2015). In fairness, CBT-based collaborative care beat usual care in primary-care anxiety (12-month response 63.7% versus 44.7%; Roy-Byrne et al., 2010) and adjunctive CBT roughly doubled response in treatment-resistant depression (46% versus 22%; Wiles et al., 2013), with a benefit still present at 3–5 years (Wiles et al., 2016); the Cochrane review rates the evidence for this as resting mostly on one trial (Ijaz et al., 2018).

Relapse and durability. CBT has an enduring effect relative to stopping medication (relapse OR 2.61 favouring CBT; no significant difference from continued medication; Cuijpers et al., 2013a; Hollon et al., 2005). Yet after acute-phase cognitive therapy ends, 29% of patients relapse within one year and 54% within two (Vittengl et al., 2007); continuation-phase CBT reduces this, especially in high-risk groups (Jarrett et al., 2001), and preventive psychological interventions cut relapse against usual care (RR 0.64; Biesheuvel-Leliefeld et al., 2015). In anxiety the follow-up record is thin: effects were not significant at 12 or more months for panic disorder, and relapse was reported in only 6 of 69 trials (van Dis et al., 2020). In a panic

trial CBT-alone response fell from 48.7% acutely to 31.9% six months after stopping (Barlow et al., 2000).

Deterioration and harm. Across guided internet-CBT trials deterioration was lower with treatment than in controls (5.8% versus 17.4%, Rozental et al., 2017; 3.36 versus 7.60, Ebert et al., 2016), but low-education patients were at higher risk and did not differ from controls in the latter. Broader unwanted effects are common and poorly recorded: 5.2% of 14,587 English therapy patients reported lasting bad effects (Crawford et al., 2016); unwanted events are expected in roughly 5–20% of patients (Linden and Schermuly-Haupt, 2014); therapists of 100 CBT patients reported side effects in 43 (Schermuly-Haupt et al., 2018); and a review of 85 trials found harmful events in more than 1 in 10 participants (Klatte et al., 2023). Only 21% of 132 psychological-intervention trials monitored harms (Jonsson et al., 2014), and none of 82 NIHR psychological trials mentioned an adverse event in the final report (Duggan et al., 2014). These are general to psychological therapy (Lilienfeld, 2007), and the CBT literature has itself called for better harm detection (Barlow, 2010). Routine outcome feedback for patients who are off track is a cheap mitigation (d = 0.23; Delgadillo et al., 2018; Lambert et al., 2018).

Reading. Dropout, non-response, relapse and harm are real, quantifiable and under-reported, so R4–R6 stand. They show that CBT leaves a large minority behind, not that it fails relative to alternatives, and almost none of the sources isolates the cognitive component.

13. Patient and context factors

Comorbidity, chronicity and adversity. Comorbid personality disorder more than doubles the odds of a poor depression outcome (OR 2.16), whatever the treatment modality (Newton-Howes et al., 2014); childhood maltreatment predicts non-response or non-remission (OR 1.43; Nanni et al., 2012); and psychotherapy for chronic depression has a small effect against controls ($d = 0.23$) and did worse than medication in direct comparisons ($d = -0.31$, attributed to dysthymia studies; Cuijpers et al., 2010b). These are prognostic factors for all treatments, not demonstrations that CBT specifically fails. For anxiety, overall diagnostic comorbidity was generally unrelated to effect size across 148 treatment samples, with diagnosis-specific exceptions (Olatunji et al., 2010). Trauma-focused CBT for PTSD in military samples shows large within-group effects (d 0.78–1.10) yet about two thirds of patients retained

the diagnosis, and the programmes were only marginally better than non-trauma-focused active controls (Steenkamp et al., 2015). Transdiagnostic CBT is one response to comorbidity (Newby et al., 2015). Baseline severity does not moderate CBT versus medication (Weitz et al., 2015), and CBT still helps medication-refractory depression (Wiles et al., 2016).

Neurodivergence, cognitive impairment, age and medical illness. The evidence on autism is thin rather than negative: ten randomized trials in autistic adults (n = 537) leave most effects very uncertain (Perez-Castilla et al., 2026), and an earlier mixed-age meta-analysis found that effects depended on who rated them ($g = 0.24$ self-report, non-significant; 0.66 informant; 0.73 clinician; Weston et al., 2016). In intellectual disability, pooled effects were small ($g = 0.43$, and 0.36 in randomized trials; Tapp et al., 2023); in dementia, CBT-family therapies had a small effect on depression (SMD −0.23) and an uncertain one on anxiety (Orgeta et al., 2022). Age itself is not a barrier: older adults recovered more often in routine IAPT data (OR 1.33; Saunders et al., 2021), and late-life anxiety CBT remitted 54% of patients, though about a third relapsed (Johnco et al., 2026). Medical comorbidity attenuates effects: CBT for chronic pain gives very small to small benefits (SMD −0.09 to −0.34; Williams et al., 2020), and insomnia CBT works but is weaker with medical than psychiatric comorbidity (Wu et al., 2015). In substance use, CBT beats minimal treatment but not other specific therapies (Magill et al., 2019). We found no meta-analysis of educational attainment or IQ as a moderator. Whether adult ADHD moderates CBT outcome is addressed in Section 11.

Culture, representation and access. Culturally adapted CBT outperforms other conditions ($g = 0.67$) and unadapted versions ($g = 0.52$; Hall et al., 2016; $d = 0.32$ in Benish et al., 2011, where adapting the explanation of the illness was the one significant moderator); depression trials show similar gains (Anik et al., 2021), and CBT transports to Chinese-descent samples with modification (Li et al., 2023). But the trial base is narrow: across 342 depression trials, linguistic minorities were far more likely to be excluded than included and effects were almost never reported by ethnic group (Polo et al., 2019), in line with the broader WEIRD-sampling critique (Henrich et al., 2010). Access shapes outcomes: in IAPT, waiting time and missed appointments predicted worse outcomes and service organisation explained 33% of the variance in reliable improvement (Clark et al., 2018); recovery

was lower in poorer areas (Delgadillo et al., 2016). In low-resource settings, lay-delivered CBT-family therapy works (adjusted mean difference −4.86 on the SSQ-14 in Zimbabwe, Chibanda et al., 2016; −7.57 on the BDI-II in India, Patel et al., 2017; OR 0.22 for major depression in Pakistan, Rahman et al., 2008), so there CBT fails through reach, not efficacy, with the pooled evidence rated mostly low certainty (van Ginneken et al., 2021; Barbui et al., 2020).

Rumination and worry. Rumination predicts the onset and course of depression and overlaps with worry (Nolen-Hoeksema et al., 2008). They are also targets: rumination-focused CBT improved residual depression in a small trial with mediation by rumination change (Watkins et al., 2011), and repetitive-negative-thinking training reduced 12-month depression and anxiety diagnoses against a waitlist (Topper et al., 2017). This suggests a process that standard restructuring may miss, not a demonstrated failure.

Reading. R7 and R8 are supported in a limited sense: patient and context factors shift outcomes, but nearly all are shared with other treatments, and the strongest CBT-specific gaps are evidence gaps (autism, cognitive impairment, minority groups) rather than measured failures.

13.1. Goals, engagement and clinical formulation

A clinically useful distinction is between reducing an unpleasant feeling and changing how a person responds to it. In ACT, acceptance and cognitive defusion—changing one's relationship to thoughts rather than treating them as literal directives—serve action guided by values (Hayes et al., 2006). The proposed target is *psychological flexibility*: persisting or changing behaviour in accordance with values and the present situation. A review of 12 mediation studies found results generally consistent with this model, but identified methodological limitations and narrow coverage of the proposed processes (Stockton et al., 2019). This is preliminary process evidence, not proof of a unique causal mechanism or superiority to CBT. It also limits a feeling-first account: valued action need not be conceptualised as waiting for distress to disappear.

A practicing clinician's perspective, shared informally with the author, distinguishes needing instructions, wanting change but being blocked by fear, resisting perceived control, and difficulty establishing collaborative engagement. We treat these as formulation questions, not a validated four-type classification, fixed traits or diagnoses. We did not identify validation of

that exact classification in our targeted searches and the readiness/reactance sources reviewed here; these searches were not exhaustive. No prevalence claims follow, and disengagement alone does not establish a psychiatric diagnosis.

Related constructs do have an empirical literature. Pretreatment readiness to change was associated with psychotherapy outcome across 76 studies and 25,917 patients (Krebs et al., 2018). That association does not establish the benefit of assigning treatment by a readiness category. A synthesis of 13 controlled studies with 1,208 patients found that people high in reactance—resistance to perceived control—did better with less directive than directive treatment (Beutler et al., 2018). This supports considering the interaction between engagement and delivery, not attributing every disagreement or poor outcome to a resistant patient. For our argument, the implication is conditional: the usefulness of an instruction or cognitive technique may depend on how it is communicated and received. It does not follow that cognitive techniques are useless.

14. Therapist, delivery and real-world factors

Therapists. Therapist effects are real but modest: about 5% of outcome variance on average (range 0.2–29%; Johns et al., 2019), 6–7% among low-intensity workers in IAPT (Firth et al., 2015b), and in one large routine dataset individual therapists' recovery rates ranged from 23.5% to 95.6% (Saxon and Barkham, 2012). Whether delivering CBT competently or adherently matters is unsettled: a meta-analysis found adherence–outcome and competence–outcome correlations near zero ($r = .02$ and .07; modalities pooled; Webb et al., 2010), while single CBT studies find competence effects mostly at the extremes or in subgroups (Branson et al., 2015; Strunk et al., 2010; Kuyken and Tsivrikos, 2008). Experience alone did not improve outcomes across 6,591 patients and 170 therapists (Goldberg et al., 2016). The therapeutic alliance includes agreement on goals and tasks and a therapeutic bond. A synthesis of 295 studies found a face-to-face alliance–outcome association of $r = .278$ (95% CI $[.256, .299]$), with a similar association for internet therapy (Flückiger et al., 2018). This is correlational and not CBT-specific. In one sample of 331 patients seen by 80 therapists, therapist-level but not patient-level alliance variation predicted outcome (Baldwin et al., 2007). Session-by-session data from 17 studies and 5,350 participants showed reciprocal relations between

early alliance and symptoms, rather than a simple one-way account (Flückiger et al., 2020).

Drift and training. Reviews argue that clinicians under-use core behavioural components and drift from "doing" to "talking" therapy (Waller, 2009; Waller and Turner, 2016); a survey found most licensed psychologists did not use exposure for PTSD (Becker et al., 2004) (dated, US). We found no pooled prevalence of drift. Training changes behaviour only when it is active and supervised: unsupervised internet training did not change competence in a randomized trial (Rakovshik et al., 2016), in line with broader reviews (Beidas and Kendall, 2010; Shafran et al., 2009).

Dose, homework and schedule. Homework compliance correlates with outcome ($r = .26$, 23 studies; Mausbach et al., 2010; $g = 0.79$ for quantity, Kazantzis et al., 2016), but patients who improve may do more homework. Improvement is curvilinear with dose, and most responders in routine care respond within 7 (low-intensity) or 14 (high-intensity) sessions (Robinson et al., 2020); session frequency may matter more than total sessions (Cuijpers et al., 2013b). Intensified exposure shortened time to response in a randomized trial; its authors reported formally equivalent symptom reduction at post-treatment and six months (Pittig et al., 2021).

Does CBT survive contact with routine care?. Largely yes. Effectiveness studies for anxiety lie within the range of efficacy trials (Stewart and Chambless, 2009), those for depression are somewhat lower with about 25% dropout (Hans and Hiller, 2013), and clinical representativeness did not erase effects (Shadish et al., 2000). IAPT shows pre–post d of 0.87 (depression) and 0.88 (anxiety) (Wakefield et al., 2021), though year-one reliable recovery was 40.3% (Gyani et al., 2013), and a small independent diagnostic follow-up found only 9.2% fully recovered (n = 90; single author; contested; Scott, 2018). Youth evidence-based practices beat usual care only modestly (mean effect 0.29; Weisz et al., 2013).

Format and personalisation. A synthesis of 20 studies with 1,418 participants found a small average difference between guided internet and face-to-face CBT ($g = 0.05$, 95% CI $[-0.09, 0.20]$), with limited evidence for individual conditions (Carlbring et al., 2018). Guided internet CBT works in routine care (Etzelmueller et al., 2020), and guided beats unguided only slightly at post-treatment (Karyotaki et al., 2021); unguided CBT has a small effect

($g = 0.27$; Karyotaki et al., 2017). A large pragmatic trial of supported computerised CBT in primary care found no benefit over usual care, with low uptake (Gilbody et al., 2015). Group and individual formats showed little average difference when treatments, patients and doses were matched (Burlingame et al., 2016). Manualisation has no direct evidence of superiority (Truijens et al., 2019); algorithm-matched care modestly beat stepped care (OR 1.40; Delgadillo et al., 2022; see also Firth et al., 2015a).

Relationship, feedback and timing. Repairing alliance ruptures was associated with better outcomes across 11 studies ($r = .29$, 95% CI $[.10, .47]$), but six training/supervision comparisons did not establish an outcome benefit ($r = .11$, $[-.09, .30]$) (Eubanks et al., 2018). A cluster-randomized trial provides more direct evidence for structured feedback: among 2,233 patients, those not on track improved more when therapists received progress feedback (Delgadillo et al., 2018). This supports monitoring and adaptation, not the claim that every informal therapeutic probe is effective.

The concern that an early reframe may feel invalidating is a plausible clinical hypothesis, but the cited alliance and reactance findings do not directly test that timing mechanism. Relational readiness must also be distinguished from whether restructuring occurs before or after exposure. In 93 participants with claustrophobic fear, restructuring before exposure and consolidation after exposure produced no detected difference in outcomes through one month (Krause et al., 2022). The cognitive exercises differed as well as their order, and the trial was not an equivalence test. The public-speaking-anxiety trial supports explicit expectancy processing before or after exposure, with a small, non-significant anxiety advantage and greater expectancy change in the after-exposure condition (Section 8). Alliance, communication and timing may condition a technique's usefulness.

Reading. R9 is partly supported. CBT works in routine care with variable yield, and some delivery changes, including structured feedback, have causal evidence. How much non-response is attributable to delivery rather than treatment content or patient/context factors remains unresolved.

15. Theory, mechanism and measurement critiques

Mechanisms are under-tested. Kazdin (2009) argues that "there is no evidence-based explanation of how or why even the most well-studied interventions

produce change", and sets standards (temporal precedence, specificity, dose–response, experiment; Kraemer et al., 2002) that most CBT mediation studies do not meet. Positive evidence exists: sudden symptom gains were preceded by cognitive change in two datasets (Tang and DeRubeis, 1999; Tang et al., 2005). It is mixed: dysfunctional-attitude change mediated symptom reduction in CBT versus interpersonal therapy, but the reverse direction fit CBT versus medication ($n = 130$; Quilty et al., 2008); the CBT-side reviews call the evidence "not yet conclusive" (Driessen and Hollon, 2010) and the evidence for skill use as a mediator preliminary (Hundt et al., 2013). Cognitive vulnerability–stress interactions predicted later depression only weakly in youth (Lakdawalla et al., 2007). Defences of the cognitive model and its updates are in Beck and Haigh (2014) and Hofmann et al. (2013).

Is the cognitive part the active part?. In a network meta-analysis of 45 trials ($n = 3{,}382$), cognitive restructuring, behavioural activation and full CBT did not differ, and all beat usual care (SMD 0.57, 0.52, 0.44) (Ciharova et al., 2021). This is consistent with a behavioural reading and equally with a cognitive one; see Section 8 for the dismantling trials. For exposure, inhibitory learning emphasises new learning through expectancy violation (Vervliet et al., 2013; Weisman and Rodebaugh, 2018). This is not necessarily an alternative to belief change: revising an expectancy can itself constitute cognitive change, even without explicit restructuring (Hofmann, 2008). A large trial found expectancy-violation-based exposure effective and optimisable (Pittig et al., 2021). Thought suppression can backfire (Wenzlaff and Wegner, 2000), and experiential avoidance gives a third-wave counter-theory (Hayes et al., 1996), though a key measure may largely track distress (Wolgast, 2014) and third-wave trials were less stringent than CBT trials (Öst, 2008). The process-based critique of one protocol per diagnosis (Hayes and Hofmann, 2021; Hofmann and Hayes, 2018) comes from advocates of the alternative.

Common factors. The alliance and therapist effects reviewed in Section 14, together with expectancy ($r = .18$; Constantino et al., 2018), are replicable and modest. Wampold (2015) argues they outweigh specific ingredients, and Enck and Zipfel (2019) frame context effects in psychotherapy; this is contested, and it explains why trials without blinding cannot cleanly isolate CBT's specific effect.

Measurement. Clinician-rated scales show larger effects than self-report ($\Delta g = 0.20$; Cuijpers et al., 2010a). Seven depression scales share little symptom

content (mean overlap 0.36; Fried, 2017); four common scales lack unidimensionality and longitudinal invariance ($n = 3{,}509$; Fried et al., 2016), and sum scores hide which symptoms change (Fried and Nesse, 2015). Response is therefore partly instrument-dependent, and standardised, patient-defined outcomes are rare (Cuijpers, 2019; Lambert and Ogles, 2009).

Evidence quality. Only 11 of 115 depression trials met eight quality criteria, with $d = 0.22$ versus 0.74 in the rest (Cuijpers et al., 2009); excess significant findings and mean power of 49% suggest small-study bias (Flint et al., 2014); only 15% of audited trials were prospectively registered (Cybulski et al., 2016); and selective reporting inflated effects (SMD -0.81 versus -0.54; Miguel et al., 2021). A Cochrane-method review rated all eight cognitive-therapy-versus-usual-care trials at high risk of bias (Jakobsen et al., 2011), though risk-of-bias tools fit psychotherapy awkwardly (Munder and Barth, 2017). Fairness requires the counterweight: a multiverse analysis of 4,281 meta-analyses still gave a mean $g = 0.56$, 90% clinically relevant, though pooling all psychotherapies (Plessen et al., 2023). Field-level critiques call for mechanism research and less redundant trials (Cuijpers, 2015; Tackett et al., 2019).

Reading. R10 and the measurement and bias part of R3 identify limits on certainty. These findings do not establish a single mechanism of CBT-specific failure. They show that causal accounts of change remain under-tested and that trial selection and measurement can inflate estimated effects.

15.1. Evolutionary learning: motivation, not a treatment ranking

The evolutionary argument begins with a defensible observation: learning from consequences does not require human verbal reasoning. Operant reward learning has been demonstrated in the mollusc *Aplysia* (Brembs et al., 2002). Comparative work also identifies conserved basal-ganglia organization across lamprey and mammals (Grillner and Robertson, 2016). Neither observation compares psychotherapies or establishes that an older mechanism produces greater clinical benefit. The prefrontal cortex is not a single recent human addition: its subdivisions include shared mammalian regions and later primate specializations (Preuss and Wise, 2022). Evolutionary conservation supplies biological plausibility, not an ordering of treatment efficacy.

The learning concepts also require separation. *Instrumental learning* concerns relations between actions and their consequences; *Pavlovian learning*

concerns predictive relations between events (Dolan and Dayan, 2013). Within instrumental control, goal-directed action is sensitive to current outcome value, whereas habitual control is relatively insensitive to outcome devaluation. These are interacting forms of control, not a literal division between a cognitive cortex and a non-cognitive basal ganglia (Dolan and Dayan, 2013). Repeating an action in a stable context can support habit formation, but repetition is not itself a separate learning system (Wood and Rünger, 2016). Five unsuccessful human habit-induction experiments caution against assuming that practice necessarily yields outcome-insensitive responding (de Wit et al., 2018). Reward-related motivation is also not identical to conscious pleasure: wanting and liking can dissociate (Berridge and Robinson, 2016).

The useful clinical hypothesis is therefore about the relation between understanding and action, not the age of a brain region. For persistent avoidance under high executive demand, structured behavioural practice may produce more durable functional change than restructuring without comparable practice. This proposed comparison concerns acquisition, generalization and cognitive cost; it is not an established treatment ranking. Nor does it require that an unpleasant feeling disappear before action changes, as the acceptance-based formulation makes explicit (Section 13.1).

The stronger claim that cognition can only consolidate experience is contradicted by laboratory evidence. Instructed reward probabilities improved decisions in one learning task (Li et al., 2011). Misleading instructions biased choices despite contrary experience in another (Doll et al., 2009). These findings show that instructions can influence learning and behaviour, not that cognitive restructuring is invariably useful in clinical care. Likewise, a behavioural procedure can change expectancies, and a cognitive procedure can prepare a patient to undertake behavioural practice. The exposure-component and timing evidence therefore constrains any universal "experience first" claim (Sections 8 and 14). The defensible research question is which combination of practice, cognitive processing and relational support works for a specified problem and patient, not which system is evolutionarily older.

16. The strongest objections

1. **CBT has a large evidence base.** Yes (Section 7). The argument concerns mechanism and moderators, not average efficacy.
2. **CBT for adult ADHD works.** It beats matched controls in medicated adults with a skills-heavy protocol; the evidence is graded low to

moderate (Section 10).

3. **Patients practise when calm.** Accepted as a plausible defence (Section 8). Rehearsal could improve cognitive skill as well as behavioural routines; it does not identify habit formation as the mechanism.
4. **Exposure is behavioural.** This is why CBT cannot be equated with cognitive restructuring. The distinction between a procedure and its mechanism matters (Section 8).
5. **Mindfulness is not better.** The comparative evidence does not establish general superiority. The hypothesis concerns when particular components help, not whether one therapy family should replace another.
6. **Cognitive therapy sometimes wins.** Yes, in social phobia (Ougrin, 2011) and in one PTSD trial (Bryant et al., 2008).
7. **The neuroscience is about acute stress.** Correct, and the human reappraisal-under-stress findings are mixed (Section 5).
8. **Executive function does not predict CBT outcome.** Largely true for overall outcome (Section 11); the argument survives only in the narrower, skill-level form.
9. **Therapists scaffold the executive load.** Plausible and untested; a therapist who generates the evidence list lowers demand and could erase a moderator effect.
10. **Evolutionary age does not rank therapies.** Correct. Conservation motivates a learning-based hypothesis but cannot establish comparative clinical benefit (Section 15.1).
11. **Cognition can help before experience.** Correct. The learning and timing evidence is reviewed in Section 15.1; acceptance-based goals are distinguished from eliminating distress in Section 13.1.
12. **Bias is not proof that efficacy is illusory.** Correct. Researcher allegiance, publication selection and reviewer preference are distinct; they apply to competing orientations too and do not establish treatment equivalence (Section 7.3).

Limitations. This is a narrative review by a non-clinician, with no independent duplicate screening. Several key studies involve older adults, children or elevated-symptom rather than diagnosed-disorder samples. Evidence from depression informs some anxiety questions, and laboratory learning is not a direct test of psychotherapy. The clinical heuristic is not a validated taxonomy,

and the added literature searches were targeted rather than exhaustive. No mechanism proposed here should be treated as an established explanation of an individual patient's outcome.

17. Predictions and a research agenda

The argument yields hypotheses, not established treatment-selection rules.

1. **P1 (skill-level moderation).** Baseline executive function predicts acquisition of restructuring skill more strongly than acquisition of exposure-based skills, but predicts overall outcome only weakly.
2. **P2 (stress-dependence).** The gap between calm and in-the-moment reappraisal is larger in low-executive-function samples and at higher arousal than under the moderate stressors studied so far.
3. **P3 (component).** In adults with ADHD and anxiety, an exposure-and-behaviour protocol is non-inferior to the same protocol plus restructuring, under a clinically justified, prespecified margin.
4. **P4 (ADHD).** ADHD status predicts less benefit from the cognitive module specifically, rather than necessarily less benefit from the full CBT package.
5. **P5 (practice).** For persistent avoidance under high executive demand, structured behavioural practice produces more durable functional improvement than restructuring without comparable practice. This prediction concerns a defined behavioural target, not all distress or every patient.

A component comparison should include restructuring, behavioural practice and their combination, with comparable therapist contact and credible rationales. Measure actual practice, expectancy change, skill acquisition, executive demand and functioning rather than inferring mechanisms from treatment labels. Acceptance-related change should include valued action despite distress, not only reductions in distress intensity. To test sequencing, hold cognitive content and contact as comparable as possible, distinguish early alliance from prior symptom change, and measure engagement alongside follow-up outcomes. **What would weaken the hypotheses:** a precisely estimated absence of the predicted executive-function interaction would weaken P1. A precisely estimated absence of a larger calm-versus-stressed reappraisal gap in low-executive-function samples would weaken P2. A clinically meaningful restructuring benefit in the specified population would contradict P3. A

precisely estimated absence of a difference in cognitive-module benefit by ADHD status would weaken P4. A precisely estimated absence of a practice advantage would weaken P5. A non-significant difference alone is insufficient: the relevant superiority or non-inferiority contrast, margin, outcome, follow-up and power must be specified before data collection. Mediation analyses also require temporal ordering and assumptions about confounding; correlated process change is not enough. The immediate research priorities are a structured umbrella search with independent screening, analyses of existing participant-level data where available, and adequately powered component/moderator designs. Any prospective clinical comparison would require ethical review and appropriate treatment provision; none is conducted here.

18. Conclusion

The ten reasons graded in Table 1 do not support the claim that CBT generally fails. They identify substantial non-response, relapse, under-recorded harms, comparator-sensitive effects, uncertain mechanisms, variable delivery and unequal access. Most of these limitations are shared with other psychological treatments. The executive-function argument remains narrower than its popular formulation: reappraisal uses working memory, but stress does not reliably abolish it, and clinical moderation remains insufficiently tested. Component evidence does not establish that explicit restructuring is always necessary, always dispensable, or the unique route to cognitive change. Evolutionary conservation motivates the learning-based hypothesis, whose clinical value requires direct testing. Acceptance, alliance, reactance and feedback research further caution against treating treatment content independently of goals and delivery. Researcher allegiance and publication selection warrant scrutiny, but do not establish CBT-specific reviewer suppression or invalidate the efficacy literature. The useful next question is which components, delivered in which way, help which patients achieve durable functional change.

Declarations

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